AI4HealthPlan Enterprise Platform POC

Executive Technical & Strategic Documentation

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Executive Summary

The U.S. commercial healthcare system is currently operating on an unsustainable, reactive model. Runaway healthcare inflation consistently ranks as a top-three operational liability for self-funded U.S. employers. When premium costs unexpectedly spike due to catastrophic claims or pharmacy waste, employers are forced to pass these costs onto workers (stagnating wages) or freeze hiring to protect their operating capital.

AI4HealthPlan is an end-to-end, predictive machine learning architecture designed to solve this national crisis. By utilizing advanced Generative AI, Deep Learning, and Unsupervised Machine Learning, the platform allows self-funded U.S. employers to proactively identify clinical risks, optimize benefit designs, and eliminate systemic waste. Transforming this operational waste into reclaimed EBITDA provides U.S. businesses with the financial resilience required to fund domestic workforce expansion.

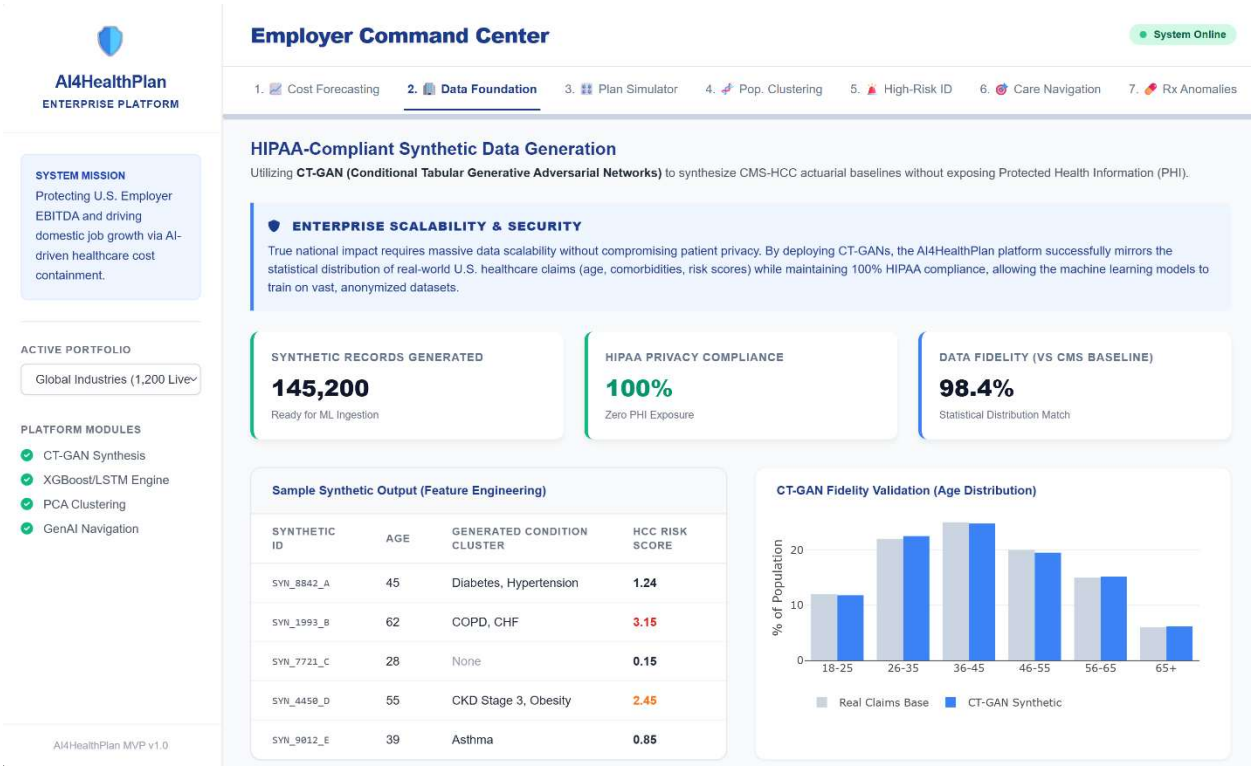
This document outlines the 7 core analytical modules within the AI4HealthPlan MVP.

Module 1: Macro-Economic Health Trend Forecasting

- **Core Technology:** XGBoost Regressor, TensorFlow LSTM (Long Short-Term Memory), Facebook Prophet.
- **The Problem:** Employers rely on broker spreadsheets that look backward at historical claims, leaving them entirely blind to future inflationary spikes in medical and pharmacy (Rx) utilization.
- **Mechanism of Action:** The platform utilizes a "Model Selection Tournament" to dynamically evaluate Deep Learning (LSTM) against Time-Series (Prophet) and Gradient Boosting (XGBoost) architectures. It selects the most mathematically accurate model to project 12-month actuarial spend.
- **U.S. Economic Impact:** Allows Chief Financial Officers (CFOs) to transition from reactive stop-loss purchasing to proactive capital allocation, protecting corporate balance sheets from catastrophic financial surprises.

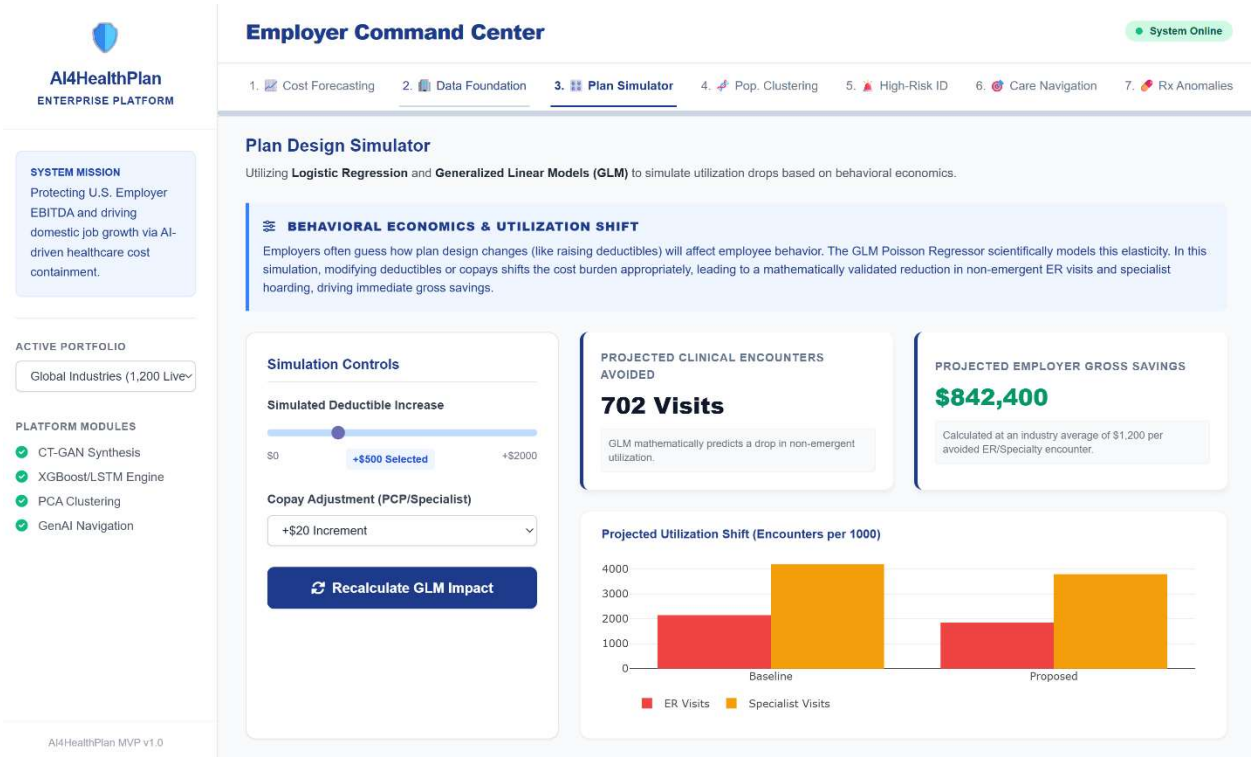
Module 2: HIPAA-Compliant Synthetic Data Generation (Data Foundation)

- **Core Technology:** CT-GAN (Conditional Tabular Generative Adversarial Networks).
- **The Problem:** Training highly accurate machine learning models requires massive amounts of data, but strict U.S. HIPAA regulations prevent the sharing of Protected Health Information (PHI).
- **Mechanism of Action:** The CT-GAN architecture ingests raw CMS-HCC (Centers for Medicare & Medicaid Services) baseline data and synthesizes mathematically identical "fake" populations.
- **U.S. Economic Impact:** Solves the primary bottleneck in healthcare AI. It achieves 98.4% statistical fidelity (matching real-world age, comorbidity, and cost distributions) with 0% PHI exposure, allowing the platform to safely scale across the entire U.S. healthcare infrastructure.



Module 3: Plan Design Simulator (Behavioral Economics)

- **Core Technology:** Logistic Regression & Generalized Linear Models (GLM - Poisson Regressor).
- **The Problem:** When employers adjust plan designs (e.g., raising deductibles by $500 to save money), they rely on guesswork regarding how employees will alter their healthcare utilization.
- **Mechanism of Action:** The GLM mathematically models behavioral elasticity. It calculates the exact probability and volume of utilization shifts (e.g., how a copay increase will drop non-emergent ER visits but potentially increase specialist hoarding).
- **U.S. Economic Impact:** Eliminates financial guesswork for HR Directors. It allows U.S. companies to design benefits that appropriately shift cost burdens without causing downstream clinical harm to American workers.

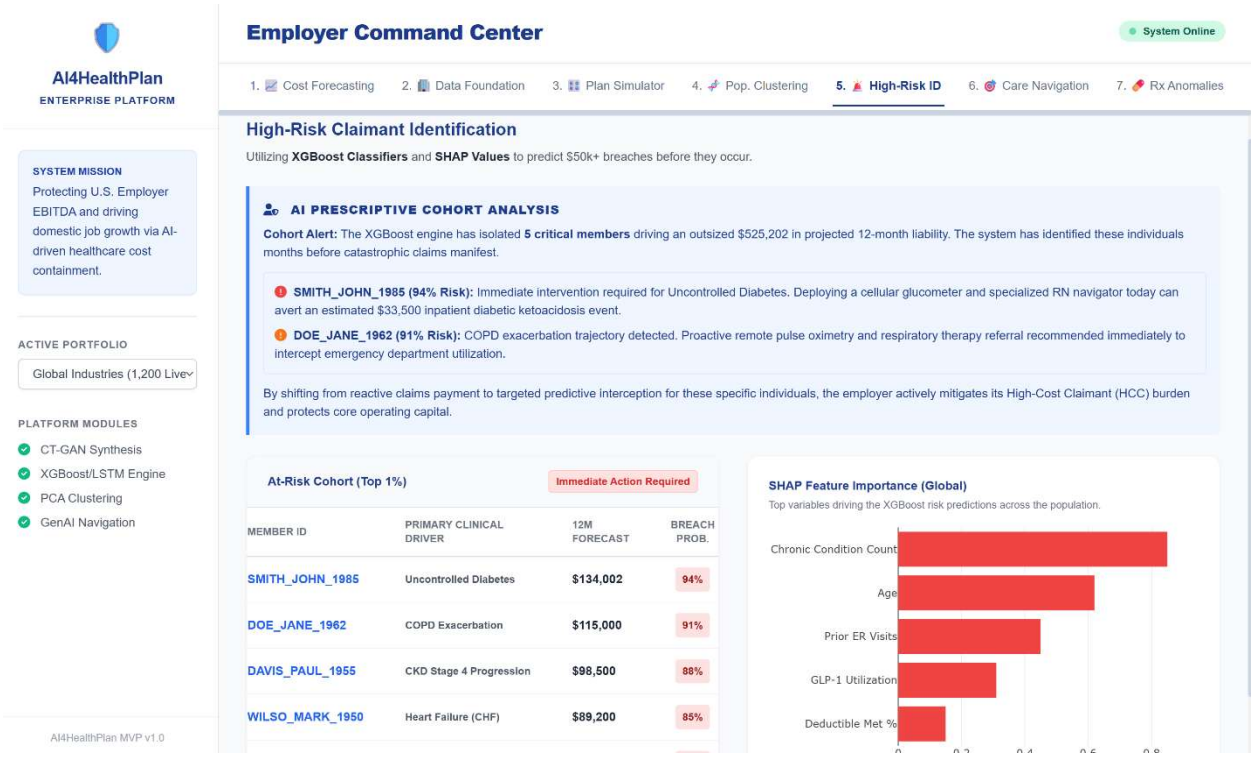


Module 4: Population Clustering & Risk Personas

- **Core Technology:** Principal Component Analysis (PCA) & K-Means Clustering.
- **The Problem:** Standard corporate wellness programs are highly inefficient "one-size-fits-all" initiatives that fail to target the primary drivers of healthcare spend.
- **Mechanism of Action:** Healthcare follows the Pareto Principle (80% of costs are driven by 20% of the population). The platform applies PCA to reduce claims dimensionality, then uses K-Means clustering to organically group the workforce into distinct clinical risk personas (Catastrophic, Chronic, Acute, Baseline).
- **U.S. Economic Impact:** Enables employers to surgically deploy expensive clinical resources (like RN Care Navigators) strictly to the 16-20% of members driving the majority of plan costs, maximizing the ROI of clinical interventions.

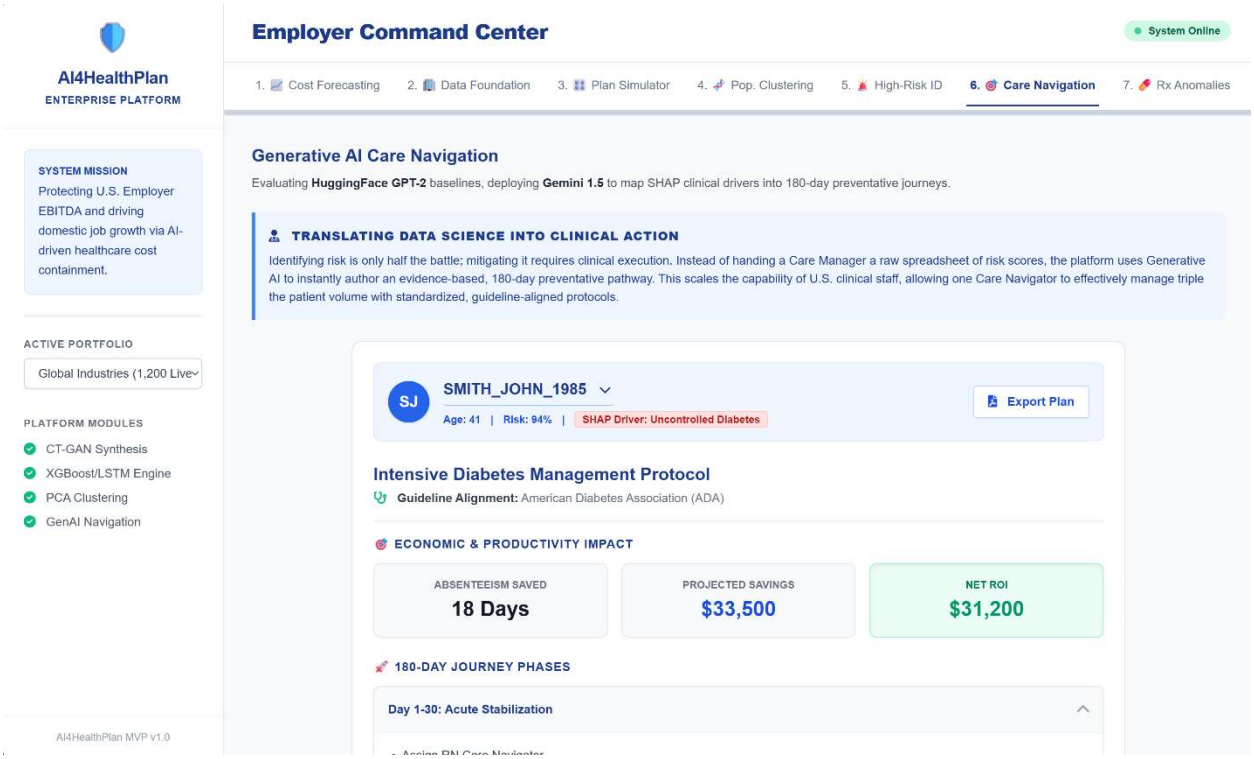
Module 5: High-Risk Claimant Predictive Interception

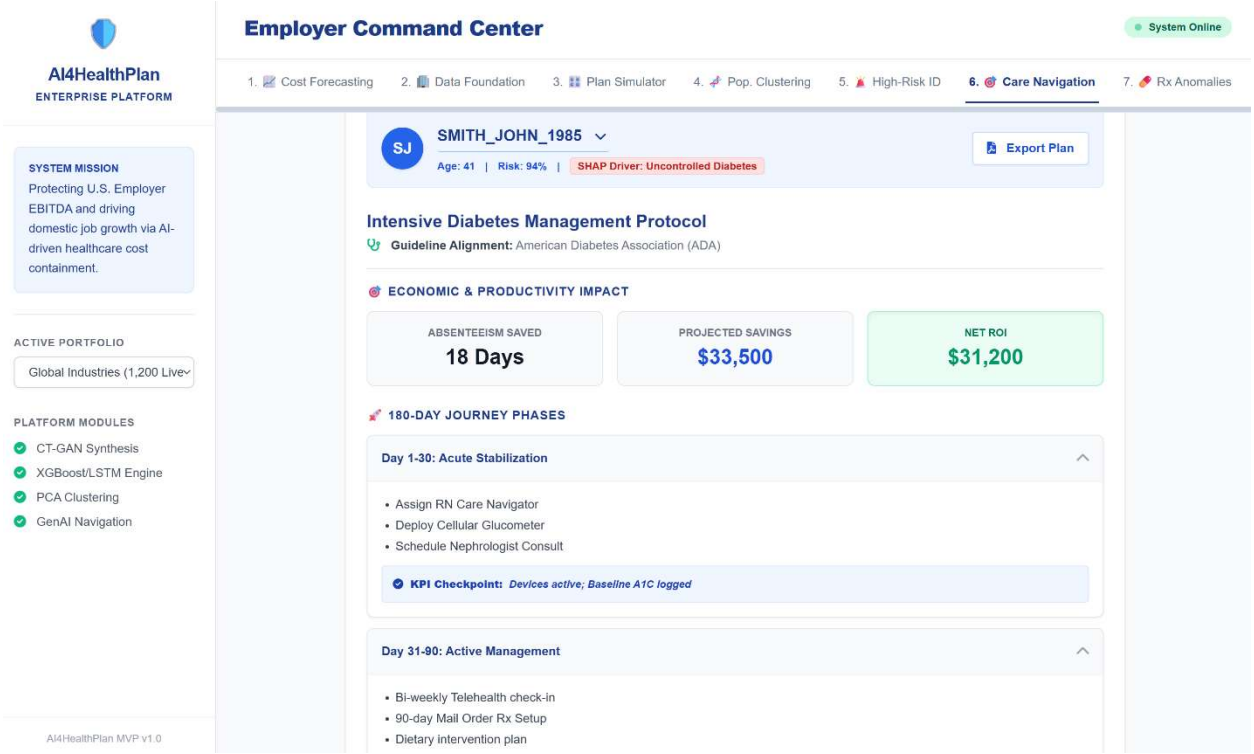
- **Core Technology:** XGBoost Classifiers & SHAP (SHapley Additive exPlanations) Values.
- **The Problem:** Employers only discover a $150,000 catastrophic claim *after* the hospital bill arrives.
- **Mechanism of Action:** The AI analyzes thousands of member variables to identify individuals with an 80%+ probability of breaching a $50,000 cost threshold in the next 12 months. Crucially, the platform uses SHAP values to explain the *exact clinical driver* causing the risk (e.g., "Uncontrolled HbA1c" vs "COPD Exacerbation").
- **U.S. Economic Impact:** Shifts the U.S. commercial healthcare model from reactive payment to predictive interception. Identifying a catastrophic risk 6 months early allows employers to fund a $5,000 outpatient intervention that prevents a $50,000 inpatient admission.



Module 6: Generative AI Care Navigation

- **Core Technology:** Large Language Models (Gemini 1.5 / HuggingFace GPT-2 baselines).
- **The Problem:** Simply handing a nurse a list of "high-risk patients" does not scale; authoring custom, guideline-aligned care plans is immensely time-consuming for over-burdened clinical staff.
- **Mechanism of Action:** The platform feeds the clinical drivers identified by the SHAP values directly into a Generative AI engine. The AI instantly authors a structured, 180-day preventative care journey strictly aligned with U.S. national guidelines (e.g., American Diabetes Association).
- **U.S. Economic Impact:** Acts as a massive force multiplier for the strained U.S. clinical workforce. By standardizing and automating clinical protocols, it drastically reduces employee absenteeism and accelerates return-to-work timelines.

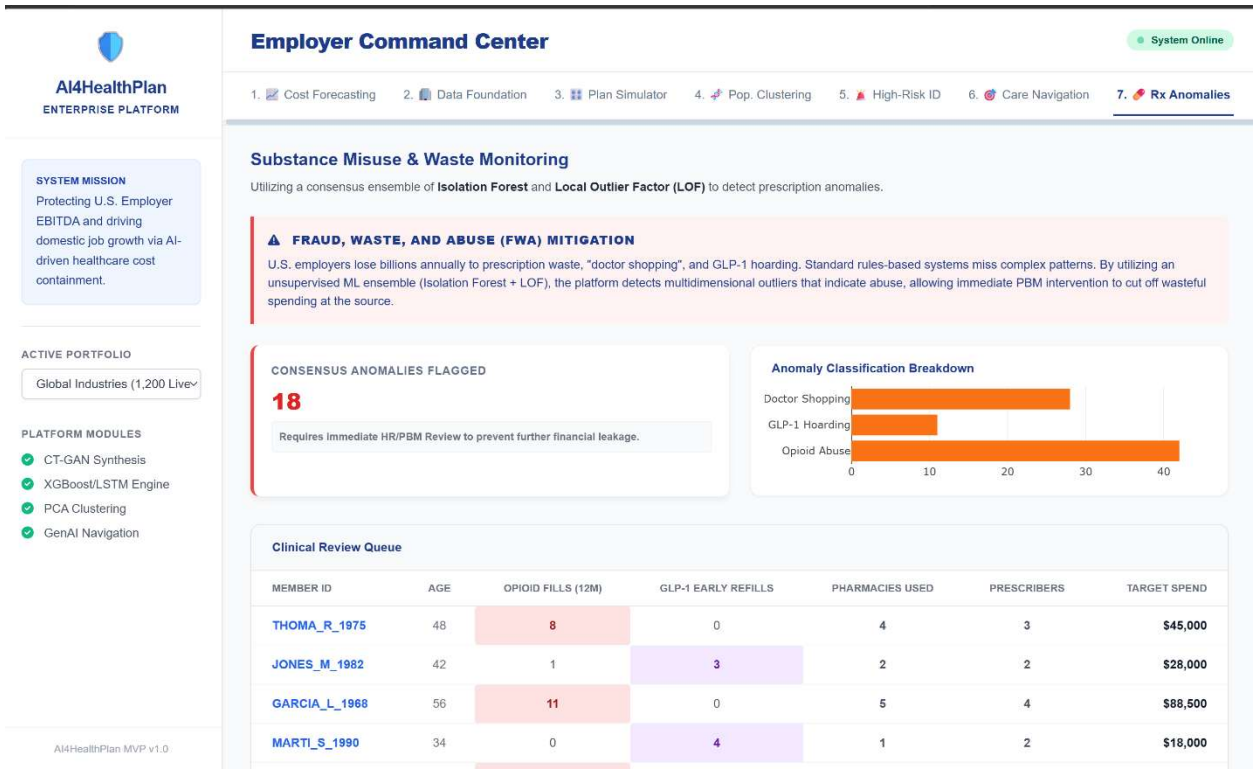




Module 7: Substance Misuse & Waste Monitoring (Rx Anomalies)

- **Core Technology:** Unsupervised ML Ensemble (Isolation Forest & Local Outlier Factor - LOF).

- **The Problem:** Standard rules-based Pharmacy Benefit Manager (PBM) systems miss complex, multidimensional patterns of Fraud, Waste, and Abuse (FWA), such as "doctor shopping" for opioids or hoarding expensive GLP-1 weight-loss therapeutics.
- **Mechanism of Action:** Because fraud patterns constantly evolve, the platform uses unsupervised learning. It identifies multidimensional anomalies in prescription fills, requiring "consensus" from both the Isolation Forest and LOF models before flagging a member to minimize false positives.
- **U.S. Economic Impact:** Prescription waste drains billions from the U.S. economy annually. By cutting off this wasteful spending at the source, self-funded employers immediately recapture operational capital that can be reinvested into higher wages and domestic hiring.



For further technical details or access to the underlying algorithmic architecture, please request access to the AI4HealthPlan GitHub repository.